water is said to be alkaline when OH– ion exceeds H+ ion and the pH value ranges between 7 to 14.

iv. Chloride

Chlorides are generally present in water in the form of Sodium chloride E.g.: Common Salt. The amount of chloride present in the water is determined by titrating the sample of water with Silver Nitrate solution taking Potassium Chromate as buffer. The end point is the appearance of red colour. The permissible limit of chloride for drinking water is 250 mg/l.

v. Residual chlorine

The free chlorine which remains as residue in treated water after the contact period is called residual chlorine. It can be determined by two methods.

a. Starch Iodide method

b. Orthotolidine Arsenite method

vi. Iron and Manganese

They usually occur together. They should be less than 0.3 ppm in portable water. They produce rust spots on fabrics (clothes) and plumbing fixtures. They are determined by chlorometric principle. Iron is determined by persulphate method. They impart reddish brown colour to water.

3. Bacteriological Tests

The following tests are normally done for bacteriological examination of water

i. Total count test

ii. E-Coli test

i. Total count test

In this test, bacterias are cultivated on specially prepared culture medium of agar containing nutrients for bacteria. The diluted sample is incubated at 20°C for 48 hours or 37°C for 24 hours. The bacterias grow and multiply and form colonies or clusters. The bacterias thus formed are counted and the results are computed for 1cc. For portable water, the total count should not exceed 100 per cc.

ii. E-Coli test

It is also called as B-Coli test. In this method, the sample of water is filtered through a sterilized membrane containing microscopic pores of size 5 to 10 millimicron to retain bacterias. The membrane with retained bacterias is incubated for 20 hours at 37°C along with



Largest water treatment plant in the world

Nearly one billion gallons of water are **processed** on an average day at the James W. Jardine Water Purification Plant in Chicago, Illinois, the largest water treatment plant in the world. This plant and the South Water Purification Plant serve nearly 5 million consumers in the City of Chicago and 118 outlying suburbs.

(1 gallon = 3.78541 litres)



Activity 2

Visit any treatment plant / pumping station available in your locality and submit a report for the functioning of the units.

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